Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
MCA | Sem-3 | Summer-2025

Course Code	:	2305CS332	**Date**	:	06-05-2025
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define Computer Network. List the advantages of computer network.	**4**
	(B)	Draw OSI layer.	**3**
		OR	
		Write a short note on fiber optic cable.	
	(C)	Define Delay. Explain types of delay.	**7**
		OR	
		Explain Network Layer with its functionality.	
Q.2	**(A)**	Explain importance of application layer in term of its protocol.	**4**
	(B)	State the port number of 1) HTTPS, 2) SMTP, 3) DNS	**3**
		OR	
		Write a short note on FTP.	
	(C)	Explain how email works using SMTP protocol.	**7**
		OR	
		Explain how DNS works with example.	
Q.3	**(A)**	Discuss Go-Back-N protocol with example.	**4**
	(B)	Draw UDP header segment.	**3**
		OR	
		Draw TCP header segment.	
	(C)	Explain rdt 2.0 with diagram.	**7**
		OR	
		Explain rdt 2.1 with diagram.	

Q.4	**(A)**	Define routing and its types.	**4**
	(B)	Find total number of host for IP address 192.168.1.1/26	**3**
		OR	
		Find total number of subnet for IP address 192.168.1.1/26	
	(C)	Explain Distance vector routing algorithm with example.	**7**
		OR	
		Explain Link state vector routing algorithm with example.	
Q.5	**(A)**	Define parity check technique with example.	**4**
	(B)	Write a short note on FDMA.	**3**
		OR	
		Write a short note on TDMA.	
	(C)	Explain CRC with example.	**7**
		OR	
		Explain pure aloha and slotted aloha.	

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